

Nate Cirino

832-833-5202 | cirino.na@northeastern.edu | portfolio | [linkedin.com/in/nate-cirino](https://www.linkedin.com/in/nate-cirino) | github.com/nateci

Graduating May 2027 | Available May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Foundations of Data Science, Algorithms & Data Structures, Computer Systems, Digital Design & Computer Architecture, Object-Oriented Design, Foundations of Software Engineering, Computer Networks, Systems Security

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Own front-of-house device stack across all Wayfair e-commerce retail brands (AllModern, Birch Lane, Joss & Main, Perigold): e-labels, mobile, and customer/associate-facing POS
- Building a supplier-agnostic e-label API unifying 4 suppliers behind one interface, powering new outlet launches
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Optimized Expert AI retrieval, search, and filtering across 20+ microservices, improving query relevance and result ranking for clinicians at scale
- Integrated Expert AI, a GenAI clinical decision support system, into CME accrual/redemption microservices, expanding LLM-backed recommendations to 250K+ clinicians
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow
- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts at scale

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Architected Flask backend with RAG retrieval pipeline serving personalized workout recommendations from Bedrock KB, driving user-facing content ranking end to end
- Evolved the retrieval stack from LangChain to Vespa to zero-ops Bedrock KB, improving recommendation relevance while reducing serving overhead
- Served 1–2M monthly model inference calls for 2K paying users at 99.99% uptime via AWS WAF, Datadog, and K8s
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows; continue as technical advisor on architecture and agentic-workflow direction

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

PROJECTS

NOAA AUV Sonar Detection | *Python, PyTorch, Deep Learning, AWS Neuron SDK, SageMaker* April 2025 – July 2025

- Trained CNN models in PyTorch on thermal sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Refined model architecture and loss formulation to address severe class imbalance across a long-tail signal distribution and sensor noise, improving detection
- Designed AWS Neuron SDK deployment path optimizing onboard custom-silicon inference under hard real-time limits

Sentinel | *LLM Agents, RAG, pgvector, FastAPI, Claude, Next.js, OpenTelemetry*

Apr. 2026 – Present

- Built an LLM agent autonomously triaging incidents via tool-use and multi-step reasoning over Prometheus, Tempo, and Loki telemetry
- Built pgvector embedding retrieval over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTables
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across multi-node Raft cluster

SKILLS & HONORS

Languages/Tools: Python, C++, Java, Rust, C, TypeScript, JavaScript, Bash, Linux, SQL, PostgreSQL, AWS, Azure, Docker, Kubernetes, Terraform, Jenkins, GraphQL

ML & Frameworks: PyTorch, TensorFlow, Pandas, RAG / Vector Retrieval (pgvector, Vespa, Bedrock KB), LangChain, FastAPI, Flask, Node.js, React, Next.js, Express

AI / Agentic Tooling: Claude Code, Claude Cowork, Cursor, Devin, GitHub Copilot, CodeRabbit

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor